Acme CU-1 control unit - Service Manual

Revision A. For trained service technicians.

1. Safety precautions

Read this section before any repair step. Safety steps always come first.

- Isolate the machine from mains power and apply lockout/tagout before opening any panel.
- Wait 5 minutes after isolation before touching inverter or capacitor terminals.
- Never bypass the door lock switch or the thermal cut-out.
- Steam and hot water lines stay pressurised after shutdown. Let them cool and vent them first.

2. Alarm codes

Code	Meaning	Condition	First action
E4	Motor inverter fault	Inverter reports over-current or over-temperature.	Isolate power, wait 5 minutes for the DC bus to discharge, inspect the inverter heat sink.
E5	Heater over-temperature	Bath temperature above 95 C.	Check the temperature probe and the heater relay.
E6	Imbalance in extraction	Drum vibration above limit at spin speed.	Redistribute the load and check the suspension dampers.
E7	Level sensor out of range	Pressure reading outside 0.2-4.8 V.	Check the sensor cable and the air tube for water.
E8	Communication timeout	No reply from the inverter for 3 seconds.	Check the bus cable between the control unit and the inverter.

3. Troubleshooting procedures

3.4 Alarm E4: Motor inverter fault

Safety: isolate the machine and apply lockout/tagout before opening the panel.

Condition: Inverter reports over-current or over-temperature.

Procedure: Isolate power, wait 5 minutes for the DC bus to discharge, inspect the inverter heat sink. If the alarm returns after the check, escalate to the service desk with the alarm code and the cycle step.

3.5 Alarm E5: Heater over-temperature

Safety: isolate the machine and apply lockout/tagout before opening the panel.

Condition: Bath temperature above 95 C.

Procedure: Check the temperature probe and the heater relay. If the alarm returns after the check, escalate to the service desk with the alarm code and the cycle step.

3.6 Alarm E6: Imbalance in extraction

Safety: isolate the machine and apply lockout/tagout before opening the panel.

Condition: Drum vibration above limit at spin speed.

Procedure: Redistribute the load and check the suspension dampers. If the alarm returns after the check, escalate to the service desk with the alarm code and the cycle step.

3.7 Alarm E7: Level sensor out of range

Safety: isolate the machine and apply lockout/tagout before opening the panel.

Condition: Pressure reading outside 0.2-4.8 V.

Procedure: Check the sensor cable and the air tube for water. If the alarm returns after the check, escalate to the service desk with the alarm code and the cycle step.

4. Maintenance schedule

Interval	Task
Daily	Clean door seal and filters
Monthly	Inspect belts and drain filter
Yearly	Test safety switches and thermal cut-outs